

PROJECT DESIGN

Visitor Management System

Introduction

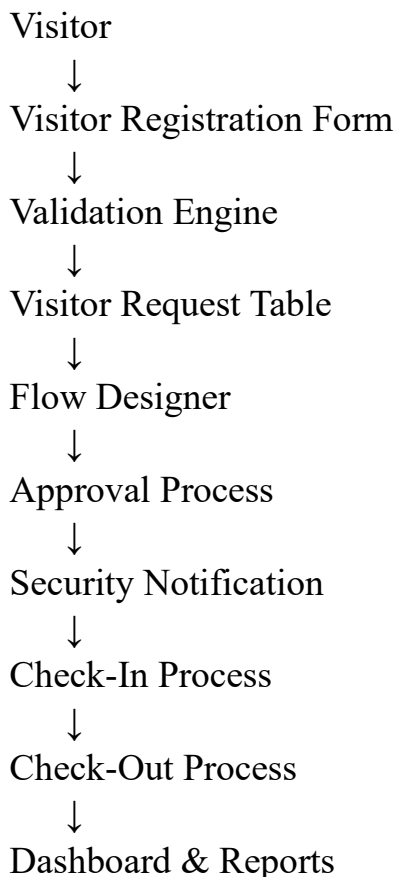
Organizations receive various types of visitors every day, including clients, vendors, interview candidates, auditors, maintenance personnel, and delivery staff. Traditional visitor management methods rely on manual registers and paper-based approvals, which can lead to security risks, delays, data loss, and inefficient tracking.

The Smart Visitor Access Management System is developed using ServiceNow to automate and streamline the complete visitor management process. The system enables visitor registration, approval workflows, blacklist verification, security notifications, visitor check-in/check-out tracking, VIP visitor management, and dashboard reporting. The solution improves security, transparency, and operational efficiency while reducing manual effort.

System Architecture Design

The application follows a layered architecture where visitor requests are submitted through a form, validated by the system, stored in a custom table, processed through approval workflows, and monitored using dashboards and reports.

System Architecture



Architecture Components

Visitor

- Submits visitor requests.

Visitor Registration Form

- Captures visitor information and visit details.

Validation Engine

- Validates phone numbers, visit dates, and duplicate records.

Visitor Request Table

- Stores all visitor-related information.

Flow Designer

- Automates approval workflows and notifications.

Security Module

- Manages visitor entry and exit tracking.

Dashboard Module

- Provides analytics and monitoring reports.

Database Design

A custom table named **Visitor Request** is created to store visitor information and manage the complete visitor lifecycle.

Table Name

Visitor Request

Fields

Field Name	Data Type	Description
Visitor ID	Auto Number	Unique Visitor Number
Visitor Name	String	Visitor Full Name
Phone Number	String	Visitor Contact Number
Email	String	Visitor Email Address
Company Name	String	Visitor Organization
Purpose	Choice	Purpose of Visit

Visit Date	Date	Planned Visit Date
Host Employee	Reference	Employee Being Visited
Status	Choice	Request Status
VIP Visitor	Boolean	VIP Indicator
Check-In Time	Date/Time	Entry Timestamp
Check-Out Time	Date/Time	Exit Timestamp
Priority	Choice	Request Priority

Form Design

The Visitor Request Form is designed to provide a simple and user-friendly interface for visitor registration.

Visitor Details Section

- Visitor Name
- Phone Number
- Email
- Company Name

Visit Details Section

- Purpose
- Visit Date
- Host Employee
- VIP Visitor

Design Considerations

- Easy navigation
- User-friendly interface
- Mandatory field validation
- Reduced data entry errors
- Improved user experience

Validation Design

To ensure data quality and security, the following validations are implemented:

Phone Number Validation

- Accepts only 10-digit phone numbers.
- Prevents invalid contact information.

Visit Date Validation

- Prevents users from selecting past dates.
- Ensures valid scheduling.

Duplicate Visitor Validation

- Detects duplicate visitor requests using phone number and visit date.
- Prevents multiple registrations for the same visitor.

Blacklisted Visitor Validation

- Checks visitor details against blacklist records.
- Blocks unauthorized visitors from registration.

Workflow Design

The workflow is designed using ServiceNow Flow Designer.

Workflow Process

Visitor Request Created



Validation Process



Blacklist Verification



Approval Request



Approved / Rejected



Security Notification



Visitor Check-In



Visitor Check-Out



Request Closed

Workflow Components

Trigger

- Activated when a new visitor request is created.

Approval Action

- Routes the request for approval.

Decision Logic

- Checks approval outcome.

Update Record

- Updates request status automatically.

Email Notification

- Sends approval information to security personnel.

Security Operations Design

Check-In Management

The system provides a Check-In button that records visitor entry time and updates the status to Checked-In.

Check-Out Management

The system provides a Check-Out button that records visitor exit time and updates the status to Checked-Out.

Security Benefits

- Real-time visitor tracking
- Accurate entry and exit records
- Improved visitor monitoring

Advanced Feature Design

VIP Visitor Management

VIP visitors are automatically assigned High Priority.

Features

- Automatic priority assignment
- VIP alert notifications
- Enhanced security monitoring

Auto Expiry Management

Visitor requests are automatically marked as Expired after the scheduled visit date.

Benefits

- Eliminates outdated records
- Improves database maintenance
- Enhances reporting accuracy

Dashboard Design

The Visitor Management Dashboard provides real-time visibility into visitor activities.

Dashboard Reports

- Today's Visitors
- Approved Visitors
- Rejected Visitors
- Checked-In Visitors
- Checked-Out Visitors
- VIP Visitors
- Expired Visitors

Dashboard Benefits

- Real-time monitoring
- Improved decision making
- Better visitor analytics
- Enhanced operational visibility

Design Advantages

Security

Ensures that only authorized visitors are allowed access to the organization.

Automation

Reduces manual approval and tracking activities.

Transparency

Provides complete visibility into visitor requests and approvals.

Scalability

Supports increasing visitor volumes without performance degradation.

User Experience

Offers a simple, intuitive, and efficient interface for users and administrators.

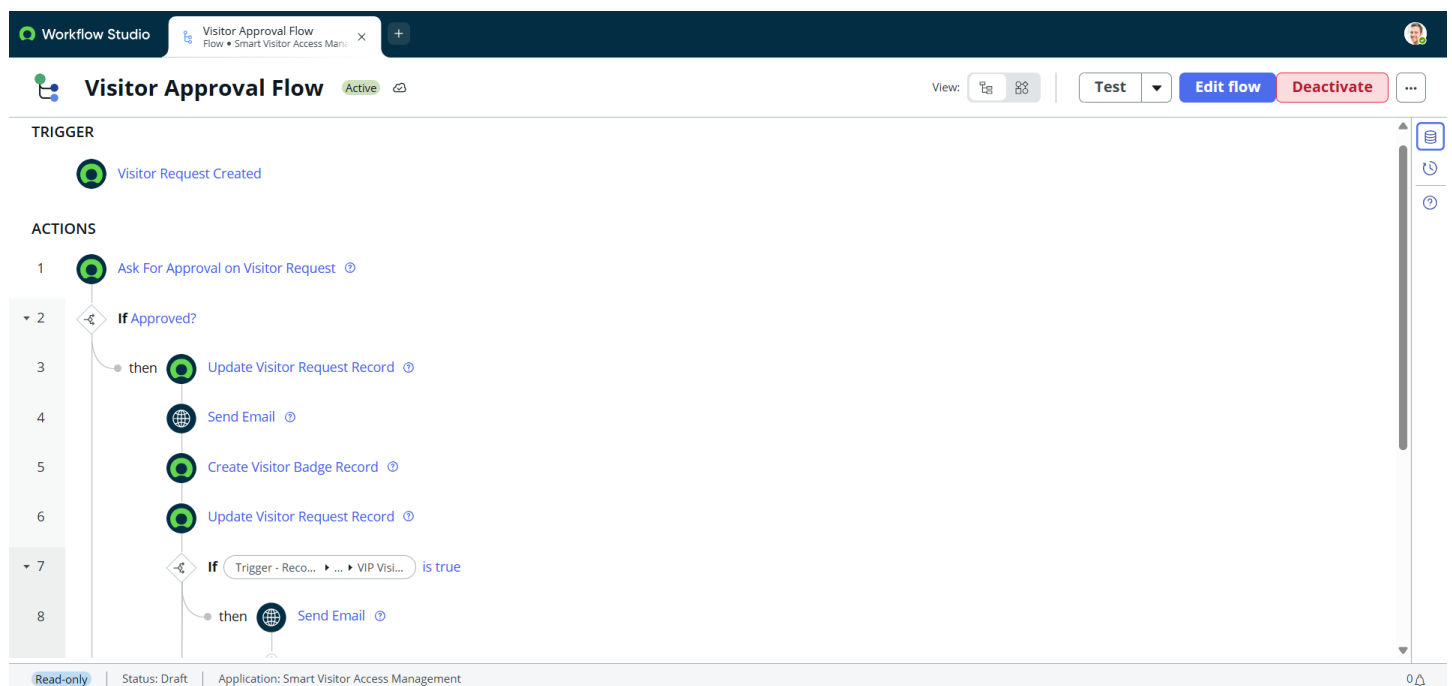
Auditability

Maintains complete records of visitor activities for future reference and compliance purposes.

Conclusion

The Smart Visitor Access Management System provides a secure, automated, and efficient approach to managing organizational visitors. The project integrates visitor registration, validation, approval workflows, security monitoring, VIP handling, automated notifications, and dashboard reporting within the ServiceNow platform. The design ensures improved security, operational efficiency, transparency, and scalability while significantly reducing manual effort and administrative workload.

Visitor Approval Workflow using ServiceNow Flow Designer



The workflow automates visitor request processing. When a visitor request is created, the system initiates an approval process. Upon approval, the visitor record is updated, email notifications are sent, visitor badge records are generated, and VIP visitors trigger additional alert notifications.